

Mijail Littin, PhD

Computational methods for inverse problems in physical systems, with regularised reconstruction and physics-informed machine learning.

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1 Current appointment

2026– **Adjunct instructor (profesor de cátedra)**, Department of Industrial Engineering, Universidad Técnica Federico Santa María (UTFSM), Chile. Four undergraduate courses (see Teaching).

2 Previous positions

2022–2025 **Doctoral researcher**, CORIA laboratory (UMR 6614 CNRS), INSA Rouen Normandie, France. RIN doctoral fellowship, Normandy Region. Collaboration with SAFRAN within the doctorate (2023–2025).

2020–2022 **Research assistant** (MSc student), Energy Conversion and Combustion Group (EC2G), UTFSM. ANID–NSFC and FONDECYT projects.

3 Education

2025 **PhD**, Normandie Université / INSA Rouen Normandie, France. Doctoral school PSIME, CORIA laboratory. Thesis: *Characterization of soot properties in laminar diffusion flames by coupling optical techniques: application to sustainable aeronautic fuels*. Defended 11 December 2025. Supervisors: J. Yon, A. Fuentes.

2024 **MSc in Industrial Engineering Sciences**, UTFSM, Chile. Thesis on soot formation in a jet-fuel laminar flame with an extended HyChem mechanism. Grade 100/100, highest distinction.

2024 **Industrial Civil Engineer**, UTFSM, Chile. Grade 100/100, highest distinction.

2020 **Licentiate in Industrial Engineering Sciences**, UTFSM, Chile.

4 Research

Inference of physical quantities that cannot be measured directly from indirect, incomplete and noisy observations, by combining physical models and data. Regularised reconstruction of ill-posed problems, including axisymmetric

tomography; joint inversion of several measurement techniques with physics-informed neural networks. Application and validation contexts: energy, combustion, thermal radiation, optical diagnostics and aerosols.

19 journal articles published or in press and 2 accepted; 7 as first author; 5 based on neural networks or machine learning. 29 conference, workshop and meeting contributions.

5 Teaching

2026-2	Design and Optimization of Thermal Systems (ILN223) , UTFSM. Diseño y Optimización de Sistemas Térmicos.
2026	Energy Management II (Heat and Mass Transfer) (ILN222) , UTFSM. Gestión Energética II (Transferencia de Calor y Masa).
2026	Energy Management I (Fluid Mechanics) (ILN221) , UTFSM. Gestión Energética I (Mecánica de Fluidos).
2026-1	Introduction to Energy Project Management (ICN367) , UTFSM. Introducción a la Gestión de Proyectos Energéticos.
2023–2025	INSA Rouen Normandie, France. Laboratory and tutorial classes in electricity, geometrical optics and wave optics (2024–2025), energy and fluids (2023–2024), and supervised student science projects (2023–2025).
2019–2022	UTFSM, Chile. Teaching assistant for graduate-level Optimization (2019–2022), Operations Research (2020), Operations Management II (2021), Fluid Mechanics (2019–2022) and Differential Calculus (2019).
2017–2018	Universidad Andrés Bello, Chile. Teaching assistant for Differential and Integral Calculus.

6 Projects

Participation in externally funded projects, without a principal-investigator role.

2025–2027	ECOS–ANID ECOS240010, DATINV: data inversion for in situ soot metrology, conventional vs neural networks. PI F. Escudero (UTFSM); French counterpart J. Yon (INSA Rouen). Co-author of the proposal, prepared and managed from the French side (2024); work on neural-network inversion.
2026	FONDECYT Regular (ANID) , PI R. Demarco, and FONDECYT de Iniciación (ANID) , PI F. Escudero. Technical support in optical diagnostics and inversion models.

2020–2022	ANID–NSFC (Chile–China): revealing soot chemistry evolution in the combustion of biofuels for jet applications. PI A. Fuentes. Research assistant.
2021–2022	FONDECYT : soot production from liquid fuels in laminar diffusion flames, application to gasoline surrogates. PI R. Demarco. Research assistant.

7 Awards and fellowships

2025	Prix Jean Bricard , Association Française d’Études et de Recherches sur les Aérosols (ASFERA), for doctoral work in aerosol science.
2022–2025	RIN doctoral fellowship, Normandy Region, France.
2022	Scientific initiation programme (PIIC), UTFSM.
2020–2024	Graduate tuition scholarship (2020–2024) and maintenance scholarship (2020–2022), UTFSM, for academic merit.

8 Publications

Journal articles, published or in press

1. **M. Littin**, F. Escudero, E. Magaña, C. López, M. Mazur, A. Fuentes, J. Yon (2026). Physics-informed neural network retrieval of spatially-resolved soot properties from multi-wavelength optical diagnostics. *Carbon* 250, 121296. doi:10.1016/j.carbon.2026.121296
2. V. Yap, J. Gallardo, **M. Littin**, F. Escudero, J. Yon, A. Fuentes, R. Demarco (2026). Dual-wave-length neural reconstruction of soot volume fraction, temperature and maturity in inverse and normal diffusion flames. *Proceedings of the Combustion Institute* 42, 106371. doi:10.1016/j.proci.2026.106371
3. M. Mercado, P. E. Pinto, I. Verdugo, **M. Littin**, F. Escudero, R. Demarco, A. Fuentes (2026). Machine-learning surrogate models for view factor estimation in benchmark fire configurations. *International Journal of Heat and Mass Transfer* 269, 129097. doi:10.1016/j.ijheatmasstransfer.2026.129097
4. P. E. Pinto, **M. Littin**, I. Verdugo, G. Severino, F. Escudero, J. J. Cruz, J. L. Urban, M. Thomsen, A. Fuentes, R. Demarco (2026). Spatially and temporally resolved soot radiation and temperature in laminar PMMA flames. *International Journal of Heat and Mass Transfer* 129542. doi:10.1016/j.ijheatmasstransfer.2026.129542
5. Y. Raynaud-Diarra, R. Ceolato, A. Poux, **M. Littin**, C. Argentin, M. Mazur, J. Yon (2026). A pair-correlation-based structure factor for fractal agglomerates accounting for primary-particle polydispersity. *Journal of Quantitative Spectroscopy and Radiative Transfer* 364, 110072. doi:10.1016/j.jqsrt.2026.110072
6. J. Yon, **M. Littin**, Y. Raynaud-Diarra, G. Lefevre, M. Mazur, M. Sztucki, R. Ceolato, A. Fuentes (2026). X-ray to visible light scattering by fractal agglomerates: a model comparison applied to soot. *Journal of Quantitative Spectroscopy and Radiative Transfer* 354, 109860. doi:10.1016/j.jqsrt.2026.109860
7. M. Weiss, **M. Littin**, F. Escudero, F. Liu, R. Demarco (2026). Assessment of extended HyChem-based gasoline kinetic mechanisms for soot production in coflow diffusion flames. *Case Studies in Thermal Engineering* 84, 108217. doi:10.1016/j.csite.2026.108217

8. P. E. Pinto, **M. Littin**, I. Verdugo, G. Severino, J. J. Cruz, F. Escudero, J. L. Urban, M. Thomsen, A. Fuentes, R. Demarco (2026). Sooting behavior of laminar flames produced with clear and black cast PMMA. *Fire Safety Journal* 163, 104875. doi:10.1016/j.firesaf.2026.104875
9. P. E. Pinto, I. Verdugo, **M. Littin**, G. Severino, J. J. Cruz, F. Escudero, R. Demarco, A. Fuentes (2026). Soot propensity of Eucalyptus globulus laminar flames at different coflow indices. *Fuel* 415, 138357. doi:10.1016/j.fuel.2026.138357
10. P. E. Pinto, **M. Littin** (2026). Improving radiation estimation from circular pool fires through field-of-view angle analysis. *International Journal of Heat and Mass Transfer* 258, 128312. doi:10.1016/j.ijheatmasstransfer.2025.128312
11. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2025). In situ determination of 2D spatially-resolved size distribution of soot aggregates in a laminar diffusion flame. *Proceedings of the Combustion Institute* 41, 105946. doi:10.1016/j.proci.2025.105946
12. **M. Littin**, M. Mazur, G. Lefevre, M. Sztucki, A. Fuentes, J. Yon (2025). Soot primary particle radial profiles in laminar diffusion flames for Jet A-1/SAF fuels: a SAXS study. *Proceedings of the Combustion Institute* 41, 105852. doi:10.1016/j.proci.2025.105852
13. G. Lefevre, G. Godard, **M. Littin**, M. Mazur, S. Richard, N. Detomaso, J. Yon (2025). Experimental investigation of the impact of organic contents and fuel (kerosene and SAF) on soot oxidation. *Combustion and Flame* 282, 114503. doi:10.1016/j.combustflame.2025.114503
14. F. Alarcón, **M. Littin**, F. Escudero, I. Verdugo, A. Fuentes, R. Demarco (2025). Mechanistic investigation of soot and NOx suppression in iso-carbon propane coflow flames by dimethyl ether substitution. *Journal of the Energy Institute* 123, 102302. doi:10.1016/j.joei.2025.102302
15. **M. Littin**, A. Poux, G. Lefevre, M. Mazur, F. Escudero, A. Fuentes, J. Yon (2024). Spline-based Abel Transform (SAT) radial property reconstruction for noise and trapping correction: application to axisymmetric sooting flames. *Fuel* 374, 132365. doi:10.1016/j.fuel.2024.132365
16. **M. Littin**, A. Poux, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). On the consideration of signal trapping for soot sizing by angular light scattering in laminar flames. *Journal of Aerosol Science* 181, 106429. doi:10.1016/j.jaerosci.2024.106429
17. **M. Littin**, F. Escudero, J. J. Cruz, I. Verdugo, D. Chen, A. Fuentes, R. Demarco (2024). Understanding soot production in a Jet A-1 laminar coflow non-premixed flame. *Proceedings of the Combustion Institute* 40, 105534. doi:10.1016/j.proci.2024.105534
18. F. Escudero, **M. Littin**, R. Demarco, F. Liu, A. Fuentes (2023). Sooting properties of laminar coflow non-premixed ethylene/hydrogen flames influenced by water vapor addition to the oxidizer. *Fire Safety Journal* 141, 103997. doi:10.1016/j.firesaf.2023.103997
19. P. Pinto, **M. Littin**, J. I. Rivera, G. Severino, J. J. Cruz, A. Fuentes (2023). A simpler tractable contour technique to model thermal radiation from buoyant diffusion flames. *Experimental Thermal and Fluid Science* 149, 111027. doi:10.1016/j.expthermflusci.2023.111027

Accepted

20. **M. Littin**, B. Kaźmierski, M. Mazur, F. Escudero, A. Fuentes, J. Yon (2026). Impact of sustainable aviation fuel on soot properties: a physics-informed neural network enhanced in-situ characterization. *Proceedings of the Combustion Institute*. Accepted.
21. N. Gutiérrez, N. Mancilla, G. Severino, **M. Littin**, J. Solís Igor, A. García, F. Escudero, J. J. Cruz, B. Herrmann, J. Yon, R. Demarco, A. Fuentes (2026). Soot maturity evolution in a periodically-forced diffusion flame via neural network assisted inversion. *Proceedings of the Combustion Institute*. Accepted.

Theses

1. **M. Littin** (2025). Characterization of soot properties in laminar diffusion flames by coupling optical techniques: application to sustainable aeronautic fuels. PhD thesis, Normandie Université / INSA Rouen Normandie (CORIA, UMR 6614 CNRS). doi:10.70675/0d4a7bfaz5f21z4970zae63zb2bfe79c5a5f
2. **M. Littin** (2024). Investigación detallada de la formación de hollín en una llama de difusión laminar de jet fuel basada en un mecanismo HyChem extendido. MSc thesis, Universidad Técnica Federico

Conference contributions (29)

1. **M. Littin**, F. Escudero, R. Demarco, M. Mazur, J. Yon (2026). Experimental determination of the radiative power emitted by a laminar diffusion flame by considering the soot optical index spatial and spectral variations. *Light Interaction with Particles / Laser-Induced Incandescence (LIP-LII 2026)*, Rouen, France.
2. F. Escudero, **M. Littin**, V. Castro, J. J. Cruz, R. Demarco, A. Fuentes, F. Liu, J. Yon (2026). Physics-based inference of soot properties from time-resolved laser-induced incandescence. *LIP-LII 2026*, Rouen, France.
3. G. Lefevre, G. Godard, **M. Littin**, M. Mazur, S. Richard, N. Detomaso, J. Yon (2026). Coupled optical diagnostics for in-situ characterization of soot oxidation: application to kerosene (Jet A-1) and sustainable aviation fuel (SAF). *LIP-LII 2026*, Rouen, France. Poster.
4. **M. Littin**, M. Mazur, G. Lefevre, M. Sztucki, A. Fuentes, J. Yon (2025). Characterization of soot emissions from aviation fuels: a SAXS study of Jet A-1 and SAF combustion aerosols. *European Aerosol Conference (EAC 2025)*, Lecce, Italy.
5. **M. Littin**, M. Mazur, G. Lefevre, M. Sztucki, A. Fuentes, J. Yon (2025). Comparison of SAXS models with the help of pair correlation functions. *11th International Symposium on Radiative Transfer (RAD-25)*, Kuşadası, Türkiye.
6. J. Yon, **M. Littin**, G. Lefevre, Y. Raynaud-Diarra, M. Mazur, A. Fuentes, M. Sztucki, R. Ceolato (2025). Structure factor of fractal aggregates based on pair correlation modeling: comparison with current modeling and impacts. *21st Electromagnetic and Light Scattering Conference (ELS XXI)*, Milazzo, Italy.
7. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2025). In situ determination of the spatially-resolved size distribution of soot aggregates in a laminar diffusion flame. *Mediterranean Combustion Symposium (MCS 2025)*, Corfu, Greece.
8. **M. Littin**, P. Pinto, I. Verdugo, G. Severino, F. Escudero, J. J. Cruz, R. Demarco, A. Fuentes (2025). Assessing the effect of wildland fuel moisture content on the spatially-resolved soot properties in laminar flames. *MCS 2025*, Corfu, Greece.
9. P. Pinto, I. Verdugo, **M. Littin**, G. Severino, J. J. Cruz, F. Escudero, R. Demarco, A. Fuentes (2025). Soot propensity of Eucalyptus globulus laminar flames at different coflow indices. *MCS 2025*, Corfu, Greece.
10. J. Yon, **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, F. Escudero, M. Sztucki (2025). Soot primary particle radial profiles in laminar diffusion flames for Jet A-1/SAF fuels: a SAXS study. *International Sooting Flame (ISF) Workshop, online meeting*.
11. **M. Littin**, M. Mazur, G. Lefevre, M. Sztucki, A. Fuentes, J. Yon (2025). Soot primary particle radial profiles in laminar diffusion flames for Jet A-1/SAF fuels: a SAXS study. *European Combustion Meeting (ECM 2025)*, Edinburgh, UK.
12. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2025). Détermination in situ de la distribution de taille des agrégats de suie dans une flamme de diffusion laminaire. *Congrès Français sur les Aérosols (CFA 2025)*, Paris, France.
13. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). Soot sizing in laminar diffusion flames: impact of self absorption. *Laser-light and Interaction with Particles (LIP 2024)*, Xi'an, China.
14. **M. Littin**, A. Poux, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). Impact of self-absorption on soot sizing by angular light scattering in a laminar axisymmetric diffusion flame. *40th International Symposium on Combustion*, Milan, Italy. Poster.
15. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). Soot sizing in laminar diffusion flame by angular light scattering. *Workshop on Measurement and Computation of Reacting Flows with Carbon Nanoparticles*, Milan, Italy. Poster.
16. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). On the consideration of self-absorption for soot sizing by angular light scattering in laminar flames. *OPTIQUE Normandie 2024*, Rouen, France.
17. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). Impact of self-absorption on soot sizing by angular light scattering in a laminar axisymmetric diffusion flame. *International Workshop on Laser-Induced Incandescence (LII 2024)*, Muskoka, Canada.
18. G. Lefevre, **M. Littin**, M. Mazur, J. Yon (2024). Impact of the soot composition and morphology on its oxidation. *LII 2024*, Muskoka, Canada. Poster.
19. G. Lefevre, **M. Littin**, J. Yon, M. Mazur (2024). Cavity-enhanced multi-wavelength light extinction and scattering measurements for drone-based soot particle characterization. *Optique et Capteurs pour l'Atmosphère (OSCA 2024)*, Toulouse, France.
20. J. Yon, M. Mazur, **M. Littin**, G. Lefevre (2024). Angular light scattering in an axisymmetric laminar flame: influence of self-absorption on soot sizing. *OSCA 2024*, Toulouse, France.

21. J. Yon, **M. Littin**, G. Lefevre, M. Bouvier, M. Mazur (2024). Diagnostic optique in-situ des particules de suie: de la flamme académique aux foyers aéronautiques. *Journée thématique de l'AFVL*, France.
 22. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2024). Correction de l'auto-absorption dans les mesures de diffusion angulaire de la lumière par les particules de suie. *Congrès Français sur les Aérosols (CFA 2024)*, Paris, France.
 23. G. Lefevre, **M. Littin**, M. Mazur, J. Yon (2024). Développement d'un banc d'essai pour la caractérisation du processus d'oxydation de nanoparticules de suie. *CFA 2024*, Paris, France.
 24. **M. Littin**, G. Lefevre, M. Mazur, A. Fuentes, J. Yon (2023). Spline-based radial property reconstruction for noise and trapping correction for emission, light scattering and extinction measurements in axisymmetric sooting flames. *GDR SUIE meeting (CNRS)*, Saint-Étienne-du-Rouvray, France.
 25. **M. Littin**, M. Mazur, A. Fuentes, J. Yon (2023). Spline-based Abel Transform (SAT) for radial property reconstruction for noise and trapping correction in axisymmetric sooting flames. *Gordon Research Conference: Laser Diagnostics in Energy and Combustion Science*, Newry, Maine, USA. Poster.
 26. **M. Littin**, R. Demarco, F. Escudero, G. Lefevre, J. Yon, M. Mazur, A. Fuentes (2023). Effects of soot coalescence process on a non-premixed Jet A-1 coflow flame using a sectional particle dynamics model. *European Combustion Meeting (ECM 2023)*, Rouen, France. Poster.
 27. **M. Littin**, J. Yon, G. Lefevre, A. Fuentes, M. Mazur (2023). Coupling extinction and angular static light scattering measurements for the characterization (2D) of soot nanoparticles in a flame. *ECM 2023*, Rouen, France. Poster.
 28. G. Lefevre, **M. Littin**, M. Mazur, J. Yon (2023). Development of a multi-wavelength optical cavity embeddable on a UAV for the characterization of soot particles in smoke plumes during industrial fires. *ECM 2023*, Rouen, France. Poster.
 29. G. Lefevre, J. Yon, A. Fuentes, **M. Littin**, M. Mazur (2023). Couplage de mesures d'extinction et de diffusion angulaire de la lumière pour la caractérisation (2D) de nanoparticules de suie dans une flamme. *Congrès Français sur les Aérosols (CFA 2023)*, Paris, France.
- Dataset: G. Lefevre, M. Littin, M. Mazur, J. Yon (2024). Combined USAXS, SAXS, WAXS techniques for studying the oxidation of soot nanoparticles in a flame. ESRF, doi:10.15151/esrf-es-1654076896 (embargoed until 2027).

9 Service

- Reviewer for *Journal of Aerosol Science, Combustion Science and Technology, Journal of Quantitative Spectroscopy and Radiative Transfer, International Journal of Heat and Mass Transfer* and *Fuel*.
- Co-organiser, Asia-Pacific Combustion Institute Summer School, UTFSM (2022 and 2025).
- Academic consulting for Lipigas, GASCO and Copec through UTFSM (2020–2025).
- Member of The Combustion Institute and ASFERA.

10 Software and tools

- **pysat-abel**: Python implementation of the spline-based Abel transform (Fuel, 2024), on PyPI since 2024.
- Programming and machine learning: Python (NumPy, SciPy), JAX, PyTorch; MATLAB, Julia, Fortran, R.
- Simulation: CoFlame (laminar reacting flows), Cantera (chemical kinetics), radiative transfer and light-scattering models.

11 Languages

Spanish (native), English (professional), French (intermediate).
